Breaking the IEEE Encryption Standard: XCB-AES in Two Queries

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CS641: Modern Cryptology
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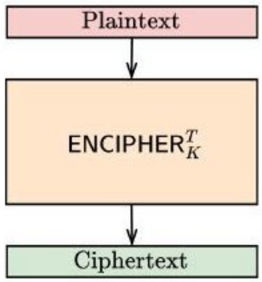
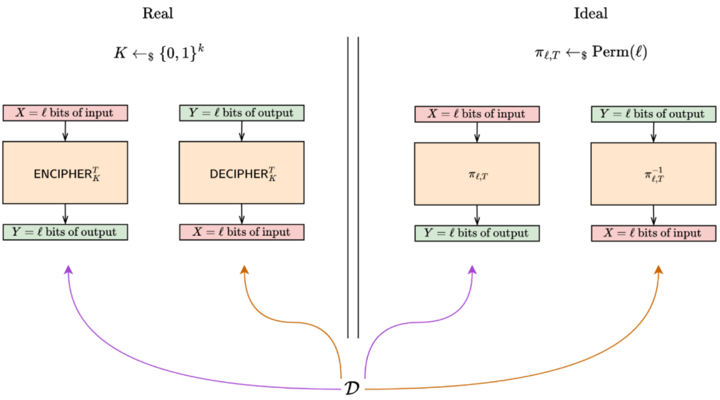
Outline

Tweakable Enciphering Modes (TEMs)

- Generalization of (tweakable) block ciphers
- Supports variable input size and tweak
- Length-preserving encryption
- Applications:
  - Disk sector and full disk encryption
  - Key wrapping and Swap-file encryption
  - Robust AEAD



Security of TEMs

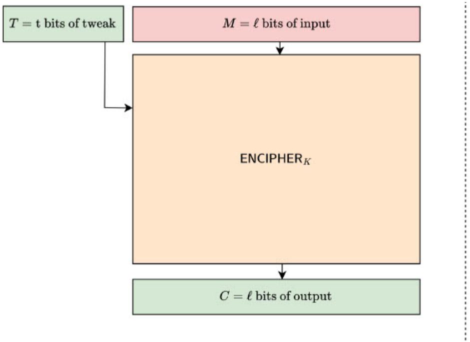


- (VIL-STPRP)Variable-Input-Length Strong Tweakable PRP
- Equivalent to strong IND-CCA style guarantees
- Ensures indistinguishability from random permutation

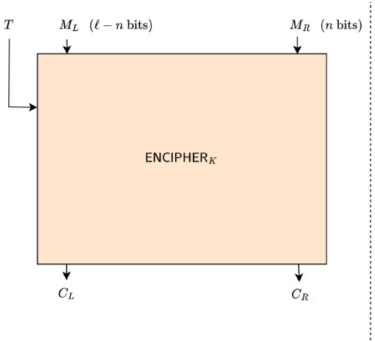
XCB-AES Standard

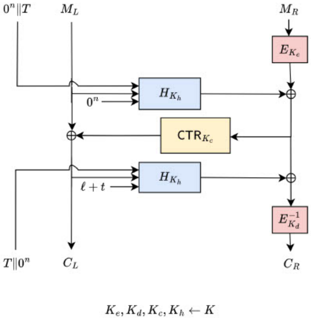
- IEEE 1619.2 standard (2010, updated 2021)
- Designed for storage encryption
- Efficient Hash-CTR-Hash construction
- XCB-AES Timeline
  - 2010: Standardized
  - 2013: Padding attack discovered
  - 2021: Updated to XCBv2 (padding-free)
  - Claimed security up to birthday bound
    - Proven VIL-STPRP up to birthday bound for block-aligned messages
    - Claimed security up to birthday bound $2^{(52-\log l)}$ Queries
  - 2024: They break XCB-AES's VIL-STPRP, STPRP and SPRP security in 2 queries. It applies to all other XCB-style modes as well

XCB-AES: IEEE 1619.2 TEM Standard



XCB-AES: IEEE 1619.2 TEM Standard

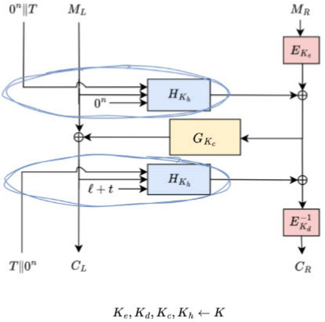




- CTR = Counter mode
- E = AES blockcipher
- H = a polynomial/rolling hash e.g., Polyval, GHASH

$$Poly_K(A_1 \| A_2 \| \dots) = A_1 K \oplus A_2 K^2 \oplus \dots$$

Their Shared Difference Attack



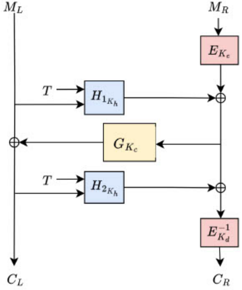
- $H_{K_h}(0^n \| T, M_L, 0^n)$

  $=\text{Poly}_{K_h}(0^n \| pad_n(T) \| Pad_n(M_L) \| 0^n)$

- $H_{K_h}(T \| 0^n, C_L, l + t)$

  $=\text{Poly}_{K_h}(pad_n(T) \| 0^n \| pad_n(C_L) \| bin_n(l + t))$

Their Shared Difference Attack



$K_e, K_d, K_c, K_h \leftarrow K$

- $H_{K_h}(0^n \| T, M_L, 0^n)$

$=\mathsf{Poly}_{K_h}(0^n \| pad_n(T) \| Pad_n(M_L) \| 0^n)$

$=\mathsf{H}_1(K_h, T, M_L)$

- $H_{K_h}(T \| 0^n, C_L, l+t)$

$=\mathsf{Poly}_{K_h}(pad_n(T) \| 0^n \| pad_n(C_L) \| bin_n(l+t))$

$=\mathsf{H}_2(K_h, T, C_L)$

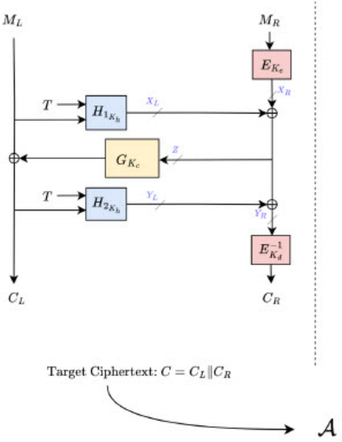
Polynomial Hash Separability

$$Poly_K(A_1 \| A_2 \| (A_3 \oplus \Delta) \| A_4) = A_1 K \oplus A_2 K^2 \oplus (A_3 \oplus \Delta) K^3 \oplus A_4 K^4$$
$$= A_1 K \oplus A_2 K^2 \oplus A_3 K^3 \oplus A_4 K^4 \oplus \Delta K^3$$
$$= Poly_K(A_1 \| A_2 \| A_3 \| A_4) \oplus Poly_K(0^n \| 0^n \| \Delta \| 0^n)$$

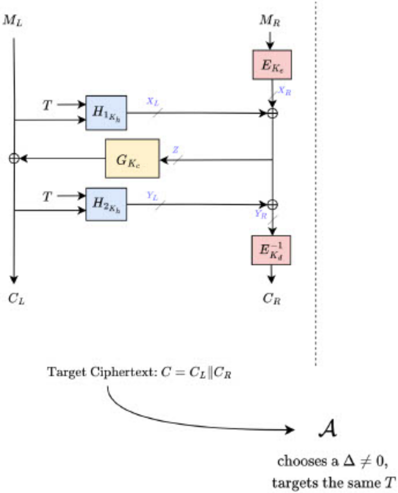
Polynomial Hash Separability

$$H_1(K_h, T, M_L \oplus \Delta) = Poly_{K_h}(0^n \| pad_n(T) \| pad_n(M_L \oplus \Delta) \| 0^n)$$
$$= \mathsf{Poly}_{K_h}(0^n \| pad_n(T) \| pad_n(M_L) \| 0^n) \oplus Poly_{K_h}(0^n \| 0^{|pad_n(T)|} \| 0^x \| \| \Delta \| 0^y \| 0^n)$$
$$= H_1(K_h, T, M_L) \oplus f_\Delta$$

$$H_2(K_h, T, C_L \oplus \Delta) = Poly_{K_h}(pad_n(T) \| 0^n \| pad_n(C_L \oplus \Delta) \| bin_n(l+t))$$
$$= \mathsf{Poly}_{K_h}(pad_n(T) \| 0^n \| pad_n(C_L) \| bin_n(l+t)) \oplus Poly_{K_h}(0^{|pad_n(T)|} \| 0^n \| 0^x \| \| \Delta \| 0^y \| 0^n)$$
$$= H_2(K_h, T, C_L) \oplus f_\Delta$$


M_L
M_R
E_{K_e}
X_R
T
H_{1K_h}
X_L
G_{K_c}
Z
T
H_{2K_h}
Y_L
Y_R
$E^{-1}_{K_d}$
C_L
C_R
Target Ciphertext: $C = C_L \| C_R$
$\mathcal{A}$

Their Shared Difference Attack



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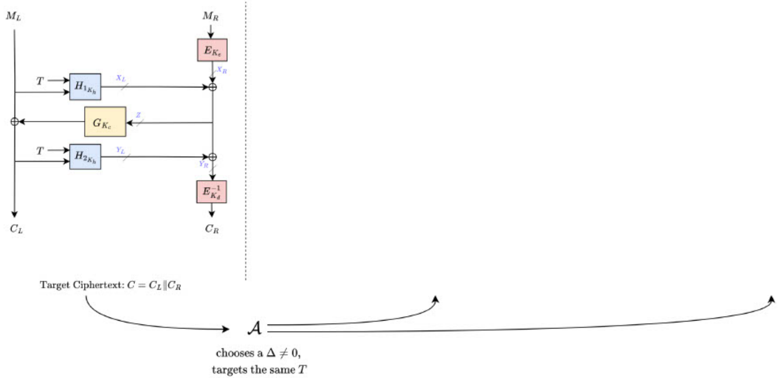

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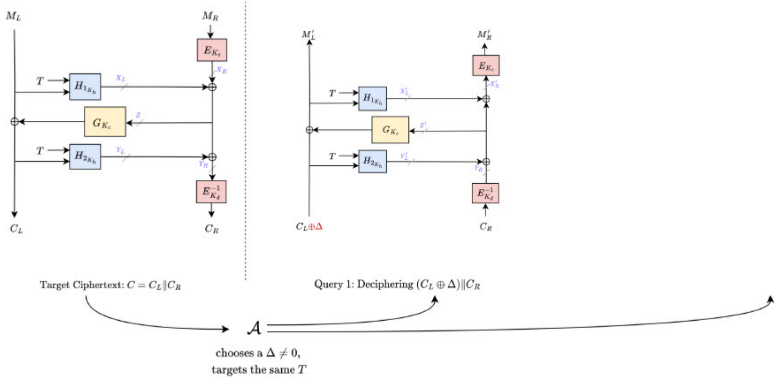

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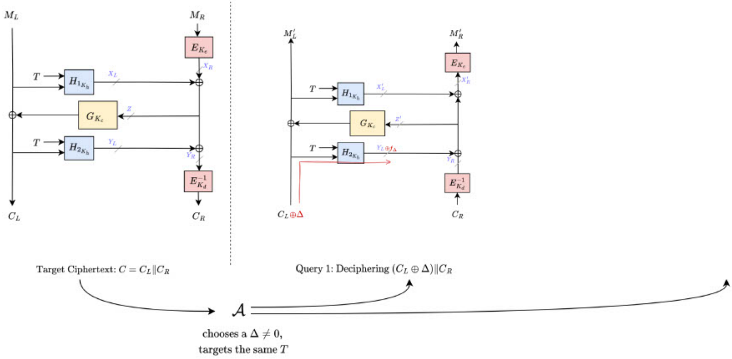

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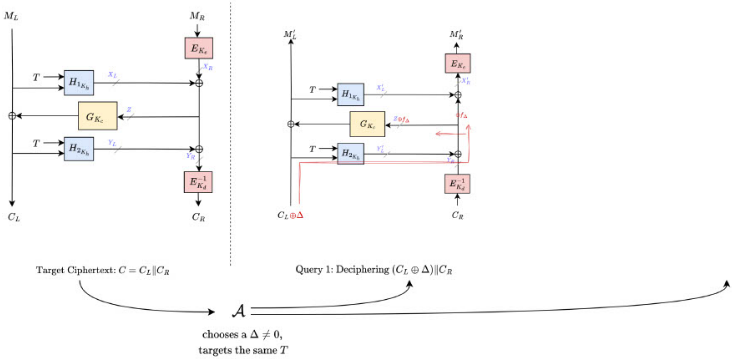

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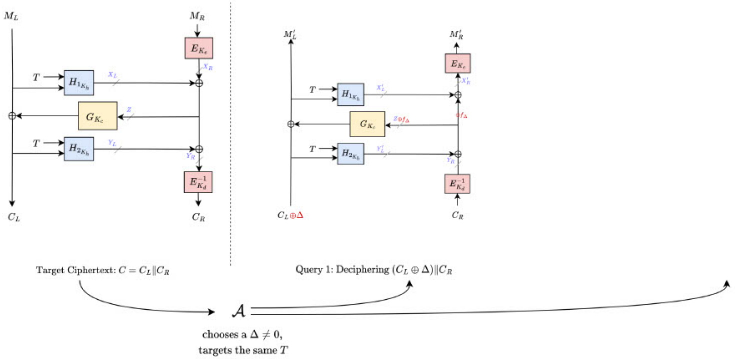

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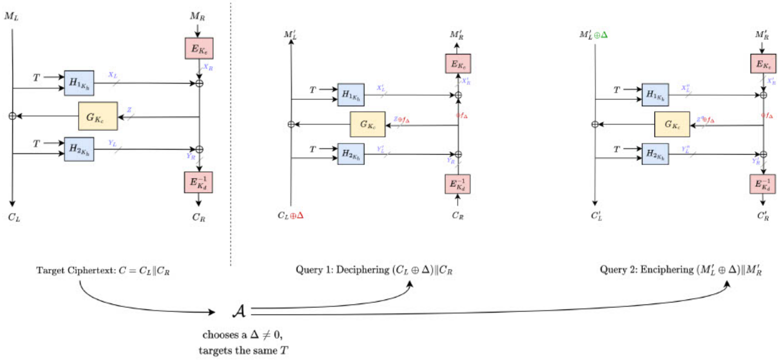

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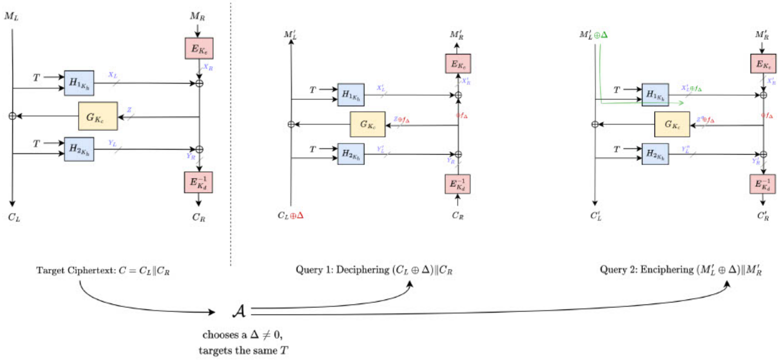

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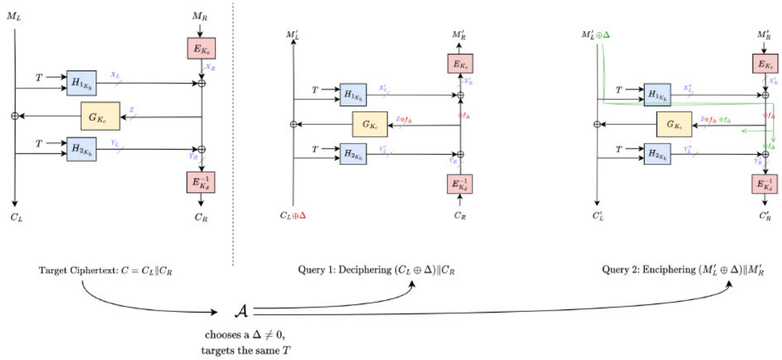

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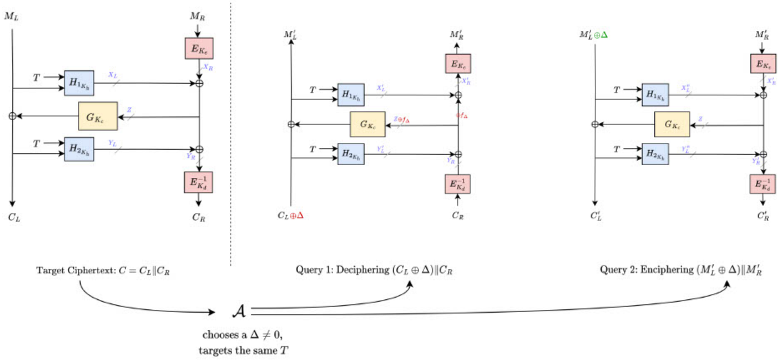

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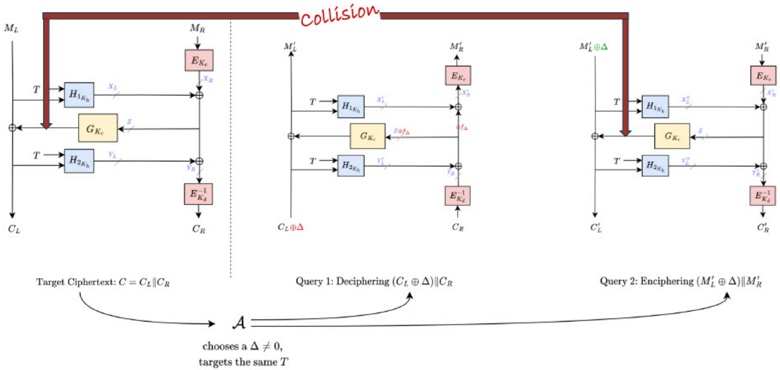

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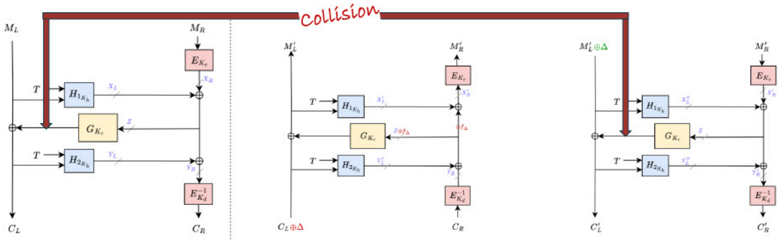
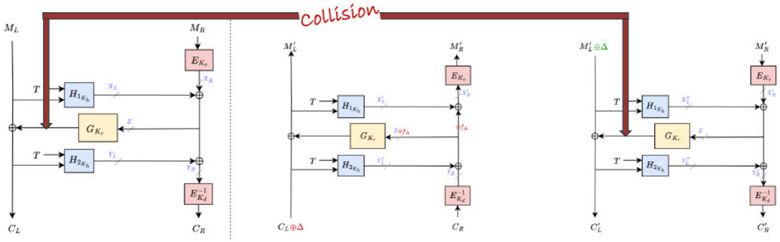


$$M_L \oplus C_L = (M_L' \oplus \Delta) \oplus C_L'$$

Their Shared Difference Attack



$$M_L \oplus C_L = (M'_L \oplus \Delta) \oplus C'_L$$
$$M_L = (M'_L \oplus \Delta) \oplus C'_L \oplus C_L$$

Message Recovered

Why the Attack Works

Root Cause: A Shared Difference Property

Shared Difference Property of Polynomial Sum

$$H_{sum}(K,T,M,C) = H_1(K,T,M) \oplus H_2(K,T,C)$$
$$H_{sum}(K,T,M \oplus \Delta, C \oplus \Delta) = H_1(K,T,M \oplus \Delta) \oplus H_2(K,T,C \oplus \Delta)$$
$$= H_1(K,T,M) \oplus f_\Delta \oplus H_2(K,T,C) \oplus f_\Delta$$
$$= H_1(K,T,M) \oplus H_2(K,T,C)$$

Due to separability of polynomial hash

$$H_{sum}(K,T,M \oplus \Delta, C \oplus \Delta) = H_{sum}(K,T,M,C)$$

Flaw in Existing Proofs

- Assumption:
  - Hash functions are XOR-universal
- Claimed:
  - Assumes H_1, H_2 are XOR-universal
  - ~~Implies H_{sum} is universal~~
  - CTR IV unpredictable and hard to collide
  - Independent and random CTR key streams up to birthday bound
- Reality:
  - Sum is **not XOR-universal**

Flaw in Existing Proofs

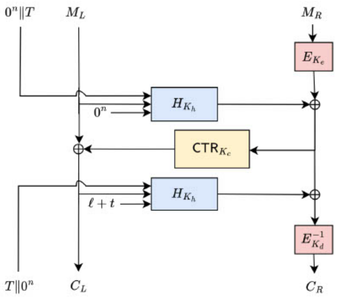
- For any two inputs $(T, M) \neq (T', M')$, output Y, and random secret key K,
  $Pr(H_1(K, T, M) \oplus H_1(K, T', M') = Y) \leq l/2^n$
  H_1 is XOR-universal
- For any two inputs $(T, C) \neq (T', C')$, output Y, and random secret key K,
  $Pr(H_2(K, T, M) \oplus H_2(K, T', M') = Y) \leq l/2^n$
  H_2 is XOR-universal
- For any two inputs $(T, M, C) \neq (T', M', C')$, output Y, and random secret key K,
  $Pr(H_{sum}(K, T, M, C) \oplus H_{sum}(K, T', M', C') = 0) = 1$
  $M' = M \oplus \Delta, C' = C \oplus \Delta$
  H_{sum} is not XOR-universal

Results Summary

Attack	Schemes	Message length	Attack type	# queries
LRW1 CCA attack by Khairallah [Kha23]	XCBv1, XCBv2, XCBv2fb	n bits	recovery of n bits	3
Shared difference attack [This work]	XCBv1	all $m > n$ bits	recovery of $m - n$ bits	4
	XCBv2, XCBv2fb, HCI, MXCB	all $m > n$ bits	recovery of $m - n$ bits	2
Flipped parts attack [This work]	HCI	all $m > n$ bits	distinguishing attack	3

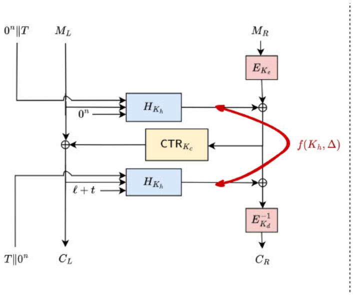
m can be arbitrarily large

Repairing and Enhancing XCB

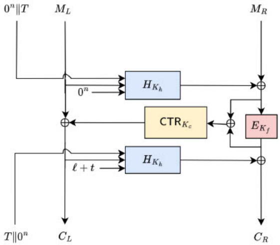


- XCB style

Repairing and Enhancing XCB



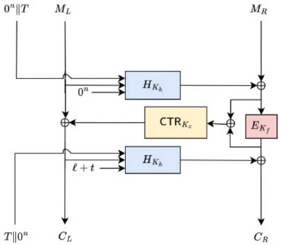
- XCB style
  - Insecure in current form.



Avoid the Sum

- XCB style
  - Insecure in current form.

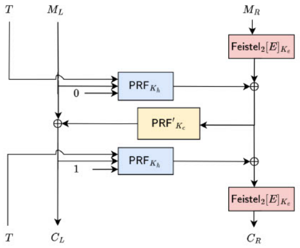
Repairing and Enhancing XCB



Avoid the Sum

- XCB style
  - Insecure in current form.
- HCTR2 style
  - AES-128, PolyVal
  - 64-bit STPRP security
  - 1.1 cpb on Gracemont

Repairing and Enhancing XCB



Use Inseparable Hashes

- XCB style
    - Insecure in current form.
- HCTR2 style
    - AES-128, PolyVal
    - 64-bit STPRP security
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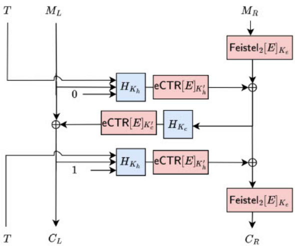
- GEM style

Repairing and Enhancing XCB



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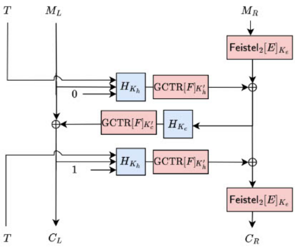
- XCB style
  - Insecure in current form.
- HCTR2 style
  - AES-128, PolyVal
  - 64-bit STPRP security
  - 1.1 cpb on Gracemont
- GEM style
  - AES-128, PolyVal
  - 128-bit STPRP security
  - 1.4 cpb on Gracemont

Repairing and Enhancing XCB



Use Inseparable Hashes

- XCB style
  - Insecure in current form.
- HCTR2 style
  - AES-128, PolyVal
  - 64-bit STPRP security
  - 1.1 cpb on Gracemont
- GEM style
  - Butterknife, PolyVal
  - 128-bit STPRP security
  - 1.1 cpb on Gracemont

Conclusion

- Their work introduced a shared difference attack against XCB
  - Breaking SPRP, STPRP, VIL-STPRP security of all XCB variants
  - Including XCB-AES; a 15 year old IEEE standard
- Pinpointed exact flaw in existing analyses
- Presented some countermeasures – HCTR2 and GEM

Impact

IEEE has officially removed XCB-AES from 1619.2 standard

Takeaway

Efforts toward TEM design and analysis through NIST's accordion initiative are essential

Thank You

Questions?